Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	Ctrl+S
Selected Domain	Domain 1 - Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	4 (primary), 8 and 9 (supporting)
Project / Solution Name	Aiversecity
Name (Leader)	Pending leader confirmation
Email & Phone (Leader)	Pending leader confirmation
University Affiliation(s)	Xiamen University Malaysia - confirm all affiliations
Number of Team Members	5 - confirm registered roster

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Fresh graduates in Malaysia who are considering data, software or operations work face a practical decision: which role should they explore, and what should they practise first? Public information often separates occupation descriptions, AI headlines and learning resources. A learner must judge whether the evidence applies to their location, distinguish job transformation from job loss, and turn that information into a realistic next step.

The ILO's 2025 global study estimates that one in four workers holds an occupation with some generative-AI exposure [1]. Exposure describes task potential, not a Malaysian displacement rate or an individual's hiring prospects. It supports the need for careful career exploration without justifying an employment prediction.

Our initial user is a university student or recent graduate comparing three supported role families. The prototype uses fictional profiles. We have not conducted user interviews or measured the prevalence of this problem in Malaysia. The immediate design hypothesis is that showing evidence, skill gaps and an ordered practice plan together will help a learner explain their next learning decision. That hypothesis needs evaluation with target users. Costs of confusion include time spent on unsuitable learning and misplaced confidence in broad AI claims, but we have not quantified those costs.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Aiversecity is a browser-based career exploration prototype. A fictional learner selects skills they have tried and answers four workplace questions. A transparent, deterministic rubric compares three response dimensions with Data Analyst, Software Developer and Operations Research Analyst profiles. It shows alignment scores, their components, self-reported evidence and missing skills. Scores describe exploration alignment rather than hiring probability.

A separate evidence layer retrieves a dated U.S. BLS employment projection. Fresh positive Data Scientists growth puts the related Data Analyst exploration first, negative growth puts it later, and zero, missing or stale evidence removes the adjustment. Personal scores keep their meaning. Each interpretation must expose its source, date, geography, related-occupation proxy and uncertainty. Refreshing evidence can therefore change a recommendation explanation or priority. Stale or insufficient evidence must remain visible. Official UN and MOSTI headlines provide context and do not serve as a demand index.

The learner can inspect a Foundation, Specialization and Capstone pathway, edit their skill profile and follow linked public learning material. Prerequisites remain in force. Every role and skill includes a proposed SDG contribution. Two qualitative AI-adoption assumptions change rubric emphasis or learning order, with explicit limits.

The running application uses rules, not a trained model or runtime LLM. Codex provides substantial development assistance and is disclosed separately. Replit is the required intended runtime. Database and GenAI workshop completion, actual platform use and sponsor requirements need confirmation. We will add a model only if it provides a justified, tested function. The current scope supports human decisions with public evidence and synthetic data.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Client: React 19.3.1 and TypeScript 5.9.3. Server: Vite 8.3.5 middleware, Node.js 22 target and fixed public-source fetches. XML parsing: fast-xml-parser 5.11.2. State: browser memory, no database or authentication. Testing: Node test runner and Playwright 1.64.1. Architecture diagram and source register follow in the appendix.

Sponsor-provided resources: no verified use recorded. Replit configuration is prepared, actual use/proof is pending. No runtime GenAI API or RunPod workload. The official portal requires API Premium sponsor integration, but sponsor identity and documentation need clarification. This is a compliance blocker.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We translated the Domain 1 journey into six connected capabilities and designed a mobile-first flow around one learning decision. The initial persona and usability assumptions come from the brief, not field research. The team still needs to validate them with target users.

Implementation separates the assessment rubric from external trend interpretation so reviewers can inspect each rule. Source timestamps, missing evidence and uncertainty belong in the interface. Synthetic practice tasks avoid real CVs and personal records.

Before Phase 2, we propose a moderated usability evaluation using fictional profiles, a timed comparison against browsing the same source material, and a comprehension check covering score meaning, evidence geography and the first pathway step. We will record denominators and failures. Target metrics in Section 4 remain goals.

Deployment preparation includes locked dependencies, documented startup commands, HTTP checks and browser journeys across screen sizes. The required Replit import, real account/project proof and independent judge-access check remain release gates. A dated fallback supports demonstrations during source outages. Final public release also requires a content and rights review, leader declarations and confirmation of unresolved official instructions.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: Target 4.4, employment-relevant technical and vocational skills. Supporting: 8.2, conditional productivity through technology, and 9.5, technology/research capability. Each contribution is proposed, not measured [2-4].
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Published context baseline: ILO 2025 reports one in four workers globally in occupations with some GenAI exposure [1]. This is neither a Malaysian rate nor a displacement estimate. Local task-time and comprehension baselines have not been measured.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Median minutes to choose and explain a role and first practice step. Comprehension pass rate: correctly identify score meaning, source geography and first prerequisite. Record paired baseline/prototype results and sample size.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed Phase 2 goals: 20% reduction in median task time relative to the same-source browsing baseline, and at least 80% comprehension pass rate in a planned small usability pilot. Both are targets, not results. Report absolute times, paired denominators and failures. No employment or SDG-indicator improvement claimed.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [ ] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops. [Pending team confirmation]
- [ ] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs. [Pending team confirmation]
- [ ] The proposed solution targets a functional, deployed application rather than a theoretical business case study. [Pending team confirmation]
- [ ] All fields in this template have been completed within the word limits stated. [Pending team confirmation]

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

Neither selected - all members must confirm pre-existing work and approvals.

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Pending member declarations, including ownership and creation dates of Figma/design assets.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

Yes - third-party libraries and public sources are used.

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React / React DOM; Vite; Tailwind; fast-xml-parser	Code	MIT
TypeScript; Playwright; Node.js	Code / runtime	Apache-2.0; Apache-2.0; MIT and bundled notices
BLS / O*NET / ILO / UN / MOSTI	Public source material	Attribution and provider terms, see source register
Figma-derived UI and fonts	Design / assets	Ownership confirmation pending; font notices required

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

Neither selected - each member must confirm employment-related IP obligations.

If yes, please describe the nature of the agreement and which member it concerns:

Pending member confirmation.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Signature: pending leader review and signature

Date: pending signature

Appendix A Architecture and references

Architecture: synthetic profile -> browser rubric -> alignment scores and skill gaps. Fixed public-source fetch -> validated dated evidence -> separate trend interpretation -> exploration or learning priority. Chosen role + skills + scenario -> prerequisite-ordered pathway -> SDG explanation. Sources never receive profile answers. No runtime model, database or GPU.

[1] ILO, Working Paper 140, 20 May 2025. https://www.ilo.org/publications/generative-ai-and-jobs-refined-global-index-occupational-exposure
[2] UN Target 4.4. https://sdgs.un.org/goals/goal4
[3] UN Target 8.2. https://sdgs.un.org/goals/goal8
[4] UN Target 9.5. https://sdgs.un.org/goals/goal9
BLS Data Scientists, U.S. long-range employment projection, related-role proxy. https://www.bls.gov/ooh/math/data-scientists.htm

Review status: prepared 23 October 2026 MYT. Pending leader facts, signatures, team review, final release links and platform proof. This review copy must not be submitted with unresolved declarations. See submission status checklist and source register.

Appendix B System block diagram

Synthetic profile and questions	Deterministic assessment rubric	Alignment scores and skill gaps
Fixed public BLS and RSS fetch	Dated evidence and eligibility rules	Exploration and practice priority
Role plus skills plus scenario	Prerequisites and public resources	Ordered pathway and SDG explanations

Each row reads left to right. The pathway combines the personal inputs with the separate eligible trend interpretation. Only the Data Analyst proxy has directional evidence. Other directions and measured skill demand remain unknown.